

SHORT COMMUNICATION

Computation of higher-order moments of generalized polynomial chaos expansions

É. Savin^{1,*} and B. Faverjon²

¹ONERA–The French Aerospace Lab, Châtillon CEDEX, F-92322, France

²INSA Lyon, Villeurbanne CEDEX, F-69621, France

SUMMARY

Because of the complexity of fluid flow solvers, non-intrusive uncertainty quantification techniques have been developed in aerodynamic simulations in order to compute the quantities of interest required in an optimization process, for example. The objective function is commonly expressed in terms of moments of these quantities, such as the mean, standard deviation, or even higher-order moments. Polynomial surrogate models based on polynomial chaos expansions have often been implemented in this respect. The original approach of uncertainty quantification using polynomial chaos is however intrusive. It is based on a Galerkin-type formulation of the model equations to derive the governing equations for the polynomial expansion coefficients. Third-order, indeed fourth-order moments of the polynomials are needed in this analysis. Besides, both intrusive and non-intrusive approaches call for their computation provided that higher-order moments of the quantities of interest need to be post-processed. In most applications, they are evaluated by Gauss quadratures and eventually stored for use throughout the computations. In this paper, analytical formulas are rather considered for the moments of the continuous polynomials of the Askey scheme, so that they can be evaluated by quadrature-free procedures instead. Matlab[®] codes have been developed for this purpose and tested by comparisons with Gauss quadratures. Copyright © 2017 John Wiley & Sons, Ltd.

Received 8 July 2016; Revised 29 November 2016; Accepted 3 January 2017

KEY WORDS: orthogonal polynomials; linearization problem; polynomial chaos; uncertainty quantification

1. INTRODUCTION

The polynomial chaos (PC), or homogeneous chaos expansion defined as the span of Hermite polynomial functionals of a Gaussian random variable, has been introduced by Wiener [1] for stochastic processes. Mean-square convergence is guaranteed by the Cameron–Martin theorem [2] and is optimal (i.e., exponential) for Gaussian processes. For arbitrary random processes, the numerical study in [3] has shown that the convergence rates are not optimal. This observation has prompted the development of generalized polynomial chaos (gPC) expansions involving other families of polynomials [3, 4]. They consist in expanding any function of random variables into a linear combination of orthogonal polynomials with respect to the probability density functions of these underlying random variables. The homogeneous and generalized homogeneous chaos expansions have recently received a broad attention in engineering sciences, where they are extensively used as a constructive tool for representing random vectors, matrices, tensors, or fields for the purpose of quantifying uncertainty in complex systems. Several applications are described in, e.g., [3–16] and references therein.

*Correspondence to: É. Savin, Computational Fluid Dynamics Department, ONERA–The French Aerospace Lab, 29, avenue de la Division Leclerc, F-92322 Châtillon CEDEX, France.

†E-mail: eric.savin@onera.fr

Complex aerodynamic analysis and design of aircraft use high-fidelity computational fluid dynamics tools for shape optimization, for example, whereby some robustness is achieved by considering uncertain operational, environmental, or manufacturing parameters. Non-intrusive uncertainty propagation is typically considered in computational fluid dynamics, because the complex flow solvers are preferably treated as black boxes in order to compute the output quantities of interest that are required to evaluate the objective function of an optimization process. The latter is often expressed in terms of moments of the quantities of interest, such as the mean, standard deviation, or even higher-order moments (skewness, kurtosis, etc.). Together with the Monte Carlo method, the method of moments, the stochastic collocation approach, or PC expansions are widely used non-intrusive approaches for evaluating stochastic objective functions. Using the latter approach, the computation of higher-order moments of the output quantities of interest in an optional post-processing step requires the evaluation of higher-order moments of the orthogonal polynomials involved in these expansions.

The intrusive approach originally introduced in [9, 15] is based on a Galerkin-type projection formulation of the model equations, typically partial differential equations, to derive the governing equations for the spectral expansion coefficients of the output quantities of interest. More precisely, the PC expansions of the model parameters and variables are substituted in the model equations, which in turn yield the evolution equations for the outputs from Galerkin projections using the orthogonal polynomials of the PC expansions. This procedure gives rise to third-order, even fourth-order moments of these polynomials as illustrated with some simple examples in [7, 9, 11]. Their computation is needed at this stage, while they may also be useful in a post-processing step if higher-order moments of the output quantities of interest are requested. The polynomial moments are evaluated by Gauss quadratures in most applications and then stored for use throughout the computations.

The main purpose of this communication is to show that analytical formulas for these moments are available, so that they could be evaluated numerically by general, quadrature-free procedures instead. A numerical implementation of these formulas is thus proposed in the form of freely available Matlab[®] codes. It is believed that such results may have some relevance for the engineering community interested in uncertainty quantification issues, using either intrusive or non-intrusive approaches based on PC and gPC expansions. The use of higher-order moments in post-processing steps, for example, is illustrated in [17] for optimization, or in [18] for global sensitivity analysis, among other possible applications. The paper is organized as follows. The standard linearization problem of a product of polynomials is very briefly introduced in the next section. Then, it is applied to the computation of higher-order moments of orthogonal polynomials in Section 3, where the available explicit expressions of the third-order moments are listed for Jacobi, generalized Hermite, and generalized Laguerre polynomials (thus covering all continuous polynomials identified in [3, 19], for example). Numerical implementation of these results using Matlab[®] is addressed in Section 4. It is validated by comparisons with classical evaluations of the third-order moments by Gauss quadratures, for which the codes used in this process are also provided. Some conclusions and perspectives are finally drawn in Section 5.

2. STANDARD LINEARIZATION PROBLEM

Let $Q_j(x)$ and $R_k(x)$ be two polynomials of degrees j and k , respectively. Let $\{P_n\}_{n \geq 0}$ be an arbitrary sequence of polynomials such that $\deg P_n = n$. The general linearization problem consists in finding the coefficients $B_n(j, k)$ such that

$$Q_j(x)R_k(x) = \sum_{n=0}^{j+k} B_n(j, k)P_n(x) . \quad (1)$$

A particular case of this problem is the standard linearization problem (or Clebsch–Gordan-type problem) for which $Q_n \equiv R_n \equiv P_n$:

$$P_j(x)P_k(x) = \sum_{n=0}^{j+k} B_n(j, k)P_n(x) . \quad (2)$$

Another particular case is the so-called connection problem, for which $R_k(x) = 1$; if in addition $Q_j(x) = x^j$ is chosen, it is referred to as the inversion problem for the sequence $\{P_n\}_{n \geq 0}$. These problems have been the subject of numerous investigations, some of them being addressed in [20–30] and references therein. The objective of this communication is definitely not to review exhaustively these results but to apply them to the computation of higher-order moments from PC or gPC expansions of random parameters and/or functionals. Such expansions have emerged as efficient numerical tools for uncertainty quantification and propagation in complex engineering systems. They have been considered in the intrusive spectral stochastic finite element method proposed in [9, 15], or in the non-intrusive stochastic collocation method proposed in [10, 16], among other possible applications.

3. HIGHER-ORDER MOMENTS OF ORTHONORMAL POLYNOMIALS

Consider now the standard linearization problem for the family $\{P_n\}_{n \geq 0}$ of orthogonal polynomials with respect to the nonnegative density $x \mapsto \mu(x)$ of support I , i.e.,

$$\langle P_m P_n \rangle_\mu := \int_I P_m(x) P_n(x) \mu(x) dx = \gamma_n \delta_{mn}, \quad m, n \geq 0, \quad (3)$$

where δ_{mn} is the usual Kronecker symbol and $\gamma_n > 0$ is the normalization constant. Then, clearly from Equation (2), the following holds:

$$\langle \hat{P}_j \hat{P}_k \hat{P}_l \rangle_\mu = \sqrt{\frac{\gamma_l}{\gamma_j \gamma_k}} B_l(j, k) := b_l(j, k), \quad l \leq j + k, \quad (4)$$

if we introduce the orthonormalized polynomials $\hat{P}_n \equiv \gamma_n^{-\frac{1}{2}} P_n$. The roles of j, k , and l in Equation (4) are transparent, so they can be permuted in this formula. This should be apparent in the analytical expression of $b_l(j, k)$ whenever it is available. In addition, one has $b_l(j, k) = 0$ whenever $l < |j - k|$. Indeed, either $k + l < j$ or $j + l < k$ in this case, thus, $\deg\{P_k P_l\} < \deg\{P_j\}$ or $\deg\{P_j P_l\} < \deg\{P_k\}$ and, consequently, $\langle P_j P_k P_l \rangle_\mu = 0$. The fourth-order moment can be derived from the aforementioned third-order moments by simple mathematical induction:

$$\begin{aligned} \langle \hat{P}_j \hat{P}_k \hat{P}_l \hat{P}_m \rangle_\mu &= \sum_{n=0}^{j+k} b_n(j, k) \langle \hat{P}_l \hat{P}_m \hat{P}_n \rangle_\mu \\ &= \sum_{n=0}^{j+k} b_n(j, k) b_n(l, m). \end{aligned} \quad (5)$$

Likewise, free permutations of the transparent indices j, k, l , and m are applicable. Higher-order moments are obtained along the same lines by repeated uses of (2) and induction. The aforementioned third-order and fourth-order moments of orthonormal polynomials typically arise in the determination of the PC expansion for the product of two or three stochastic variables, as illustrated in [7], for example. Here, the third-order tensor $C_{jkl} = \langle P_j P_k P_l \rangle_\mu$ is rather evaluated numerically by dedicated quadrature rules, benefiting to some extent from its sparsity. As explained in the introductory section, these moments are needed in the spectral stochastic finite element method; for example, the PC expansions for model parameters and variables are substituted into the governing equations; then, using a Galerkin projection method, evolution equations are obtained for the spectral coefficients in the PC expansions. In non-intrusive stochastic collocation methods, the spectral coefficients of the PC expansions of the output quantities of interest are computed by running the underlying physical model for particular model parameter values, typically belonging to an adapted quadrature set [13]. Both in the intrusive and non-intrusive methods, the moments are used to post-process the PC expansions for deriving the moments of the output quantities of interest, so long as they are needed.

The linearization coefficients for some classical families of orthogonal polynomials are explicitly given in the subsequent sections. Jacobi, generalized Hermite, and generalized Laguerre polynomials are more particularly addressed. Families corresponding to discrete nonnegative measures $\mu(dx)$ may be considered alike, although they are not reviewed in this communication.

3.1. Jacobi polynomials

The Jacobi polynomials $\{J_n^{(\alpha,\beta)}\}_{n \geq 0}$ are orthogonal with respect to the weight function $\mu(x) = (1-x)^\alpha(1+x)^\beta$, with $\alpha, \beta > -1$ and $I = [-1, 1]$. They are defined by, e.g., the standard Rodrigues' formula:

$$J_n^{(\alpha,\beta)}(x) = \frac{1}{\mu(x)} \frac{d^n}{dx^n} \left(\frac{\mu(x)}{n!} \left(\frac{x^2-1}{2} \right)^n \right) = \sum_{j=0}^n \binom{n+\alpha}{n-j} \binom{n+\beta}{j} \left(\frac{x-1}{2} \right)^j \left(\frac{x+1}{2} \right)^{n-j},$$

where

$$\binom{z}{p} = \frac{z!}{(z-p)!p!}$$

stands for the generalized binomial coefficient. Indeed, one has $z! := \Gamma(z+1) = \int_0^{+\infty} t^z e^{-t} dt$ and $\Gamma(p+1) = p!$, the usual factorial, if p is an integer. Jacobi polynomials arise in gPC expansions for random variables following beta distributions of the first kind; see, e.g., [3, 13]. The normalization constant γ_n in (3) then reads

$$\gamma_n = \frac{2^{\alpha+\beta+1} (n+\alpha)! (n+\beta)!}{(2n+\alpha+\beta+1) (n+\alpha+\beta)! n!}.$$

The linearization coefficients $B_n(j, k)$ in the general linearization problem

$$J_j^{(\lambda,\delta)}(x) J_k^{(\mu,\nu)}(x) = \sum_{n=|j-k|}^{j+k} B_n(j, k) J_n^{(\alpha,\beta)}(x), \quad \alpha, \beta, \lambda, \delta, \mu, \nu > -1,$$

are given in [23, equation (12)] in terms of double hypergeometric functions (the so-called Kampé de Fériet functions). In the context of PC expansions, we are rather interested in the standard linearization problem for which $\alpha = \lambda = \mu$ and $\beta = \delta = \nu$. A representation in terms of generalized hypergeometric series was derived in [20, 28] for this problem. For numerical robustness, we will rather resort to the older induction formula derived in [26]. Here, the linearization coefficients are given by

$$B_n(j, k) = (-1)^{j+k+n} \frac{k!}{(k+\beta)!} \tilde{B}_n(j, k), \quad (6)$$

where the coefficients $\tilde{B}_n(j, k)$ are obtained by the induction formula [27, equation (4.13)]:

$$\begin{aligned} & \frac{[(j+k+\alpha+\beta+1)^2 - (n+\alpha+\beta)^2][(n+\alpha+\beta)^2 - (j-k)^2]}{(2n+\alpha+\beta)[2(n-1)+\alpha+\beta+1]} (n+\beta) \tilde{B}_{n-1}(j, k) \\ & - \frac{[(j+k+\alpha+\beta+1)^2 - (n+1)^2][(n+1)^2 - (j-k)^2]}{[2(n+1)+\alpha+\beta][2(n+1)+\alpha+\beta+1]} (n+1+\alpha) \tilde{B}_{n+1}(j, k) \\ & + \frac{[(j+k+\alpha+\beta+1)^2 - (n+1+\alpha+\beta)^2][(n+1)^2 - (j-k)^2]}{2(n+1)+\alpha+\beta} \left(\frac{\beta-\alpha}{2} \right) \tilde{B}_n(j, k) \\ & - \frac{[(j+k+\alpha+\beta+1)^2 - (n+\alpha+\beta)^2][n^2 - (j-k)^2]}{2n+\alpha+\beta} \left(\frac{\beta-\alpha}{2} \right) \tilde{B}_n(j, k) = 0, \end{aligned}$$

starting from (assuming $j \geq k$):

$$\begin{aligned} \tilde{B}_{j-k-1}(j, k) &= 0, \\ \tilde{B}_{j-k}(j, k) &= \frac{[2(j-k)+\alpha+\beta+1]!(2k+\alpha+\beta)!(j+\alpha)!(j+\beta)!}{(2j+\alpha+\beta+1)!(k+\alpha+\beta)!(j-k+\alpha)!(j-k)!}. \end{aligned}$$

The ultraspherical (Gegenbauer) polynomials $\{C_n^{(\lambda)}\}_{n \geq 0}$ correspond to the particular case $\alpha = \beta = \lambda - \frac{1}{2}$ with $\lambda \neq 0$ and the standardization

$$C_n^{(\lambda)}(x) = \frac{(2\lambda)_n}{\left(\lambda + \frac{1}{2}\right)_n} J_n^{(\lambda-\frac{1}{2}, \lambda-\frac{1}{2})}(x), \quad \lambda > -\frac{1}{2},$$

where $(z)_n := \frac{\Gamma(z+n)}{\Gamma(z)}$ stands for the usual Pochhammer symbol. The corresponding linearization coefficients $B_n(j, k)$ such that

$$C_j^{(\lambda)}(x)C_k^{(\lambda)}(x) = \sum_{n=0}^{\min(j,k)} B_{j+k-2n}(j, k) C_{j+k-2n}^{(\lambda)}(x), \quad \lambda > -\frac{1}{2},$$

are given by Dougall's formula [31, equation (5.7)] (see also [23, equation (28)]):

$$B_{j+k-2n}(j, k) = \frac{(\lambda + j + k - 2n)(j + k - 2n)!}{(\lambda + j + k - n)n!(j - n)!(k - n)!} \frac{(2\lambda)_{j+k-n}(\lambda)_{j-n}(\lambda)_{k-n}(\lambda)_n}{(2\lambda)_{j+k-2n}(\lambda)_{j+k-n}}. \quad (7)$$

For the family of Legendre polynomials such that $\lambda = \frac{1}{2}$, the Neumann–Adams formula [21, p. 91; 32] is recovered, namely,

$$B_{j+k-2n}(j, k) = \frac{2(j + k - 2n) + 1}{2(j + k - n) + 1} \frac{(j + k - n)! \left(\frac{1}{2}\right)_{j-n} \left(\frac{1}{2}\right)_{k-n} \left(\frac{1}{2}\right)_n}{\left(\frac{1}{2}\right)_{j+k-n} (j - n)!(k - n)!n!}, \quad (8)$$

where $\left(\frac{1}{2}\right)_n = \frac{(2n)!}{4^n n!}$, and others. Legendre polynomials arise in gPC expansions for the important case of uniform distributions.

Lastly, Chebyshev polynomials of the first kind $\{T_n\}_{n \geq 0}$ correspond to the special case $\alpha = \beta = -\frac{1}{2}$ and are

$$T_n(x) = \cos(n \arccos x) = \frac{J_n^{(-\frac{1}{2}, -\frac{1}{2})}(x)}{J_n^{(-\frac{1}{2}, -\frac{1}{2})}(1)}, \quad J_n^{(-\frac{1}{2}, -\frac{1}{2})}(1) = \frac{1}{n!} \left(\frac{1}{2}\right)_n.$$

Because

$$2T_j(x)T_k(x) = T_{|j-k|}(x) + T_{j+k}(x),$$

the linearization coefficients are simply $B_{|j-k|}(j, k) = B_{j+k}(j, k) = \frac{1}{2}$ and $B_n(j, k) = 0$ otherwise.

3.2. Hermite polynomials

The generalized Hermite polynomials $\{H_n^{(\alpha)}\}_{n \geq 0}$ are orthogonal with respect to the weight function $\mu(|x|)$ with $\mu(x) = x^{2\alpha} e^{-x^2}$, $\alpha > -\frac{1}{2}$, on $I = \mathbb{R}$. They are given by the Rodrigues-like formula [19, p. 157]:

$$H_n^{(\alpha)}(x) = \frac{1}{\mu(x)} \frac{d^n}{dx^n} \left(\mu(x) x^n K_n^{(\alpha)}(x) \right), \quad (9)$$

where

$$K_{2m}^{(\alpha)}(x) = \frac{(-1)^m}{(\alpha + 1)_m} {}_1F_1 \left(\begin{matrix} m \\ m + \alpha + 1 \end{matrix} \middle| x^2 \right),$$

$$K_{2m+1}^{(\alpha)}(x) = \frac{(-1)^m}{(\alpha + 1)_{m+1}} x {}_1F_1 \left(\begin{matrix} m + 1 \\ m + \alpha + 2 \end{matrix} \middle| x^2 \right),$$

and ${}_pF_q$ is the generalized hypergeometric function defined as

$${}_p f_q \left(\begin{matrix} (a_p) \\ (b_q) \end{matrix} \middle| x \right) = \sum_{k=0}^{\infty} \frac{(a_1)_k (a_2)_k \cdots (a_p)_k}{(b_1)_k (b_2)_k \cdots (b_q)_k} \frac{x^k}{k!}.$$

The normalization constant γ_n in Equation (3) reads [19, p. 157]

$$\gamma_n = 2^{2n} \left\lfloor \frac{n}{2} \right\rfloor! \left(\left\lfloor \frac{n+1}{2} \right\rfloor + \alpha - \frac{1}{2} \right)!,$$

where $\lfloor \cdot \rfloor$ is the largest integer function. This family reduces to the classical Hermite polynomials $\{H_n\}_{n \geq 0}$ for $\alpha = 0$. The latter arises in PC expansions for random variables following Gaussian distributions and is the original PCs of the stochastic finite element method introduced in [9, 15]. Rodrigues' formula (9) for $\alpha = 0$ and $\mu(x) = e^{-x^2}$ reads

$$H_n(x) = \frac{1}{\mu(x)} \frac{d^n}{dx^n} ((-1)^n \mu(x)) = n! \sum_{j=0}^{\lfloor \frac{n}{2} \rfloor} \frac{(-1)^j (2x)^{n-2j}}{(n-2j)! j!},$$

and the normalization constant is (owing to $(-\frac{1}{2})! = \sqrt{\pi}$)

$$\gamma_n = \sqrt{\pi} 2^n n!.$$

The linearization coefficients $B_n(j, k)$ in the general linearization problem

$$H_j^{(\lambda)}(x) H_k^{(\mu)}(x) = \sum_{n=0}^{\min(j, k)} B_{j+k-2n}(j, k) H_{j+k-2n}^{(\alpha)}(x), \quad \alpha, \lambda, \mu > -\frac{1}{2},$$

are given in [22, equation (3.23)] and [24, equation (3.5)] for the standardization of generalized Hermite polynomials $\{\mathcal{H}_n^{(\alpha)}\}_{n \geq 0}$ introduced by Rosenblum [33]:

$$\mathcal{H}_n^{(\alpha)}(x) = \frac{n!}{\gamma_\alpha(n)} H_n^{(\alpha)}(x),$$

where $n \mapsto \gamma_\alpha(n)$ plays the role of a generalized factorial

$$\begin{aligned} \gamma_\alpha(2m) &= 2^{2m} m! \left(\alpha + \frac{1}{2} \right)_m, \\ \gamma_\alpha(2m+1) &= 2^{2m+1} m! \left(\alpha + \frac{1}{2} \right)_{m+1}. \end{aligned}$$

Again, in the context of PC expansions, we are rather interested in the standard linearization problem $\lambda = \mu = \alpha$, for which the linearization coefficients for the chosen standardization (9) read

$$B_{j+k-2n}(j, k) = \frac{\gamma_\alpha(j) \gamma_\alpha(k)}{\gamma_\alpha(j+k-2n)} \sum_{p=0}^{\lfloor \frac{j}{2} \rfloor} \sum_{q=0}^{\lfloor \frac{k}{2} \rfloor} \frac{\gamma_\alpha(j+k-2(p+q))}{\gamma_\alpha(j-2p) \gamma_\alpha(k-2q) p! q!} \frac{(-n)_{p+q}}{n!}. \quad (10)$$

The explicit linearization formula for classical Hermite polynomials $\alpha = 0$ is known as the Feldheim's formula and reads [25] (see also [24, equation (3.10)]):

$$H_j(x) H_k(x) = \sum_{n=0}^{\min(j, k)} \binom{j}{n} \binom{k}{n} 2^n n! H_{j+k-2n}(x), \quad (11)$$

where $\binom{j}{n} = \frac{j!}{(j-n)! n!}$ is the usual binomial coefficient for two integers $j \geq n$. We arrive at

$$B_l(j, k) = \frac{\sqrt{\pi} 2^n j! k!}{(n-j)!(n-k)!(n-l)!}$$

whenever $2n = j + k + l$ is even, and $l \leq j + k$, $k \leq j + l$, $j \leq k + l$; and $B_l(j, k) = 0$ otherwise. This formula agrees with, e.g., [34, p. 273] up to a proper normalization of the Hermite polynomials.

3.3. Laguerre polynomials

The generalized Laguerre polynomials $\{L_n^{(\alpha)}\}_{n \geq 0}$ are orthogonal with respect to the weight function $\mu(x) = x^\alpha e^{-x}$, with $\alpha > -1$ and $I = [0, +\infty)$. They are defined by, e.g., the Rodrigues' formula:

$$L_n^{(\alpha)}(x) = \frac{1}{\mu(x)} \frac{d^n}{dx^n} \left(\mu(x) \frac{x^n}{n!} \right) = \sum_{j=0}^n \binom{n+\alpha}{n-j} \frac{(-x)^j}{j!}$$

and arise in gPC expansions for random variables following gamma distributions; see, e.g., [3, 14]. This family reduces to the classical Laguerre polynomials $\{L_n\}_{n \geq 0}$ for $\alpha = 0$, applicable to exponentially distributed random variables. The normalization constant γ_n in (3) reads

$$\gamma_n = \frac{(n+\alpha)!}{n!}.$$

The linearization coefficients $B_n(j, k)$ in the general linearization problem

$$L_j^{(\lambda)}(x) L_k^{(\mu)}(x) = \sum_{n=|j-k|}^{j+k} B_n(j, k) L_n^{(\alpha)}(x), \quad \alpha, \lambda, \mu > -1,$$

are given in [22, equation (3.24)] in terms of double hypergeometric functions. Again, in the context of PC expansions, we are rather interested in the standard linearization problem $\lambda = \mu = \alpha$, for which the linearization coefficients are given by [29, 30] in terms of a terminating hypergeometric series ${}_3F_2$:

$$B_{j+k-n}(j, k) = \frac{(-2)^n}{n!} \frac{(j+k-n)!}{(j-n)!(k-n)!} {}_3F_2 \left(\begin{matrix} -\frac{n}{2}, -\frac{n-1}{2}, j+k-n+\alpha+1 \\ j-n+1, k-n+1 \end{matrix} \middle| 1 \right).$$

The first $\max(n-j, n-k)$ terms of the series earlier are ignored whenever $n > \max(j, k)$; thus,

$$\begin{aligned} B_{j+k-n}(j, k) &= \frac{(-2)^n (j+k-n)!}{(j+k-n+\alpha)! n!} \\ &\times \sum_{l=\max(0, n-j, n-k)}^{\lfloor \frac{n}{2} \rfloor} \left(-\frac{n}{2} \right)_l \left(-\frac{n-1}{2} \right)_l \frac{(j+k-n+\alpha+l)!}{(j-n+l)!(k-n+l)!}. \end{aligned} \quad (12)$$

4. NUMERICAL IMPLEMENTATION

The various formulas earlier have been implemented in Matlab[®]. The routines are distributed under CeCILL-C license and are freely available at <https://github.com/ericsavin/LinCoef/>. They were compared with the results obtained with classical Gauss quadratures for the computation of the third-order moments of (4). The Golub–Welsch algorithm [35] is used for computing Gauss quadrature weights and nodes. The recurrence coefficients for monic Jacobi, generalized Hermite, and generalized Laguerre polynomials in

$$P_{n+1}(x) = (x - c_n)P_n(x) - d_n P_{n-1}(x)$$

are given in Table I, together with the leading-order coefficient p_n and the zero-th moment $\mu_0 = \langle 1 \rangle_\mu$ for completeness. Computations by the analytical formulas detailed in the foregoing section are in very good agreement with Gauss quadratures, which validate our proposed codes.

The main function is `LinCoef.m`, which computes the linearization coefficients for Jacobi, Gegenbauer, generalized Hermite, and generalized Laguerre polynomials of arbitrary parameters

Table I. Recurrence coefficients for monic Jacobi, generalized Hermite, and generalized Laguerre polynomials.

	c_n	d_n	p_n	μ_0
$J_n^{(\alpha,\beta)}(x)$	$\frac{\beta^2 - \alpha^2}{(2n + \alpha + \beta)(2n + \alpha + \beta + 2)}$	$\frac{4n(n + \alpha)(n + \beta)(n + \alpha + \beta)}{(2n + \alpha + \beta)^2[(2n + \alpha + \beta)^2 - 1]}$	$\frac{1}{2^n} \binom{2n + \alpha + \beta}{n}$	$2^{\alpha + \beta + 1} \frac{\alpha! \beta!}{(\alpha + \beta + 1)!}$
$H_n^{(\alpha)}(x)$	0	$\frac{1}{2}[n + \alpha(1 - (-1)^n)]$	2^n	$(\alpha - \frac{1}{2})!$
$L_n^{(\alpha)}(x)$	$2n + \alpha + 1$	$n(n + \alpha)$	$\frac{(-1)^n}{n!}$	$\alpha!$

α and β . Chebyshev polynomials (Jacobi polynomials with $\alpha = \beta = -\frac{1}{2}$) are also specifically addressed. Three routines are provided to compare the implementation with Gauss quadratures: `TestHermite.m`, `TestJacobi.m`, and `TestLaguerre.m`. These quadrature sets are constructed with the `GNodeWt.m` function, while the polynomials are evaluated at the quadrature nodes by the dedicated functions `PGHern.m`, `PJacn.m`, and `PGLagn.m`.

5. CONCLUSIONS

In this paper, we have presented the existing results for the computation of the so-called linearization coefficients for products of orthogonal polynomials of the Jacobi, generalized Hermite, and generalized Laguerre families. These coefficients not only correspond to the third-order moments of orthogonal polynomials, but they also serve for the computation of higher-order moments by induction. Therefore, they can be used in the intrusive and non-intrusive PC expansion methods for uncertainty quantification of engineering systems, among other possible applications. In the intrusive approach, the third-order or fourth-order moments arise from Galerkin-type projections of the governing equations of the system models and are needed to carry out the overall uncertainty quantification (UQ) analysis. In both intrusive and non-intrusive approaches, these moments pertain to the computation of higher-order moments (skewness, kurtosis, and beyond) of the output quantities of interest in a post-processing step, so long as they are needed. These results have been implemented in Matlab[®], and the codes have been validated by comparison with usual Gauss quadratures. The present overview concerns continuous polynomials, but it can be extended to discrete polynomials alike.

ACKNOWLEDGMENTS

The work of É.S. has been partially supported by the European Union's Seventh Framework Programme for research, technological development, and demonstration under grant agreement #ACP3-GA-2013-605036 (UMRIDA Project www.umrida.eu).

REFERENCES

- Wiener N. The homogeneous chaos. *American Journal of Mathematics* 1938; **60**(4):897–936.
- Cameron R, Martin W. The orthogonal development of nonlinear functionals in series of Fourier–Hermite functionals. *Annals of Mathematics* 1947; **48**(2):385–392.
- Xiu D, Karniadakis GE. The Wiener–Askey polynomial chaos for stochastic differential equations. *SIAM Journal of Scientific Computing* 2002; **24**(2):619–644.
- Soize C, Ghanem RG. Physical systems with random uncertainties: chaos representations with arbitrary probability measure. *SIAM Journal of Scientific Computing* 2004; **26**(2):395–410.
- Clouteau D, Savin É, Aubry D. Stochastic simulations in dynamic soil–structure interaction. *Meccanica* 2001; **36**(4):379–399.
- Creamer DB. On using polynomial chaos for modeling uncertainty in acoustic propagation. *The Journal of the Acoustical Society of America* 2006; **119**(4):1979–1994.
- Debusschere BJ, Najm HN, Pèbay PP, Knio OM, Ghanem RG, Maître Le OP. Numerical challenges in the use of representations for stochastic processes. *SIAM Journal of Scientific Computing* 2005; **26**(2):698–719.
- Ghanem RG. Ingredients for a general purpose stochastic finite elements implementation. *Computer Methods in Applied Mechanics and Engineering* 1999; **168**(1–4):19–34.

9. Ghanem RG, Spanos PD. *Stochastic Finite Elements: A Spectral Approach*. Springer: New York NY, 1991.
10. Ghiocel DM, Ghanem RG. Stochastic finite-element analysis of seismic soil–structure interaction. *ASCE Journal of Engineering Mechanics* 2002; **128**(1):66–77.
11. Najm HN. Uncertainty quantification and polynomial chaos techniques in computational fluid dynamics. *Annual Review of Fluid Mechanics* 2009; **41**:35–52.
12. Maître Le OP, Knio OM. *Spectral Methods for Uncertainty Quantification. With Applications to Computational Fluid Dynamics*. Springer: Dordrecht, 2010.
13. Savin É, Resmini A, Peter J. Sparse polynomial surrogates for aerodynamic computations with random inputs, 2016. AIAA Paper #2016-0433.
14. Segui B, Faverjon B, Jacquet-Richardet G. Effects of random stiffness variations in multistage rotors using the polynomial chaos expansion. *Journal of Sound and Vibration* 2013; **332**(18):4178–4192.
15. Sun T-C. A finite element method for random differential equations with random coefficients. *Journal on Numerical Analysis* 1979; **16**(6):1019–1035.
16. Xiu D, Hesthaven JS. High-order collocation methods for differential equations with random inputs. *SIAM Journal on Scientific Computing* 2005; **27**(3):1118–1139.
17. Singh T, Singla P, Konda U. Polynomial chaos based design of robust input shapers. *ASME Journal of Dynamic Systems, Measurement, and Control* 2010; **132**(5):051010, 13. available at <http://dx.doi.org/10.1186/s13662-015-0509-4>.
18. Geraci G, Congedo PM, Abgrall R, Iaccarino G. High-order statistics in global sensitivity analysis: decomposition and model reduction. *Computer Methods in Applied Mechanics and Engineering* 2016; **301**:80–115.
19. Chihara TS. *An Introduction to Orthogonal Polynomials*. Gordon and Breach: New York NY, 1978.
20. Abd-Elhameed WM. New formulas for the linearization coefficients of some nonsymmetric Jacobi polynomials. *Advances in Difference Equations* 2015; **2015**:168, 13. available at <http://dx.doi.org/10.1155/1.4001793>.
21. Adams JC. On the expression of the product of any two Legendre's coefficients by means of a series of Legendre's coefficients. *Proceedings of the Royal Society of London* 1878; **27**:63–71.
22. Chaggara H. Operational rules and a generalized Hermite polynomials. *Journal of Mathematical Analysis and Applications* 2007; **332**(1):11–21.
23. Chaggara H, Koepf W. On linearization coefficients of Jacobi polynomials. *Applied Mathematics Letters* 2010; **23**(5):609–614.
24. Chaggara H, Koepf W. On linearization and connection coefficients for generalized Hermite polynomials. *Journal of Computational and Applied Mathematics* 2011; **236**(1):65–73.
25. Feldheim E. Quelques nouvelles relations pour les polynômes d'Hermite. *Journal London Mathematical Society* 1938; **13**(1):22–29.
26. Gasper G. Linearization of the product of Jacobi polynomials I. *Canadian Journal of Mathematics* 1970; **22**(1):171–175.
27. Hylleraas EA. Linearization of products of Jacobi polynomials. *Mathematica Scandinavica* 1962; **10**:189–200.
28. Rahman M. A non-negative representation of the linearization coefficients of the product of Jacobi polynomials. *Canadian Journal of Mathematics* 1981; **33**(4):915–928.
29. Sánchez-Ruiz J, Artés PL, Martínez-Finkelshtein A, Dehesa JS. General linearization formulae for products of continuous hypergeometric-type polynomials. *Journal of Physics A: Mathematical and General* 1999; **32**(42):7345–7366.
30. Watson GN. A note on the polynomials of Hermite and Laguerre. *Journal of the London Mathematical Society* 1938; **13**(1):29–32.
31. Askey R. Orthogonal polynomials and special functions. In *CBMS-NSF Regional Conference Series in Applied Mathematics*, vol. 21 SIAM: Philadelphia PA, 1975.
32. Neumann FE. *Beiträge Zur Theorie Der Kugelfunctionen II*. Teubner: Leipzig, 1878.
33. Rosenblum M. Generalized Hermite polynomials and the Bose-like oscillator calculus. In *Operator Theory: Advances and Applications*, Ball JA, Dym H, Kaashoek MA, Langer H, Tretter C (eds), vol. 73 Birkhäuser Verlag: Basel; 369–396, 1994.
34. Sullivan TJ. *Introduction to Uncertainty Quantification*. Springer: Berlin, 2016.
35. Golub GH, Welsch JH. Calculation of Gauss quadrature rules. *Mathematics of Computation* 1969; **23**(106):221–230.